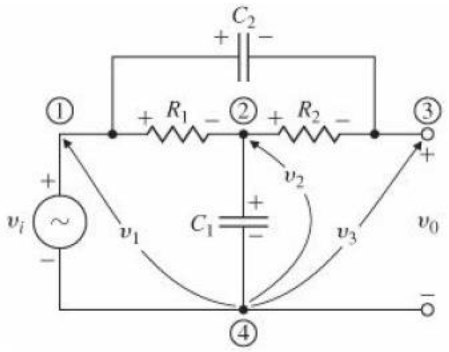
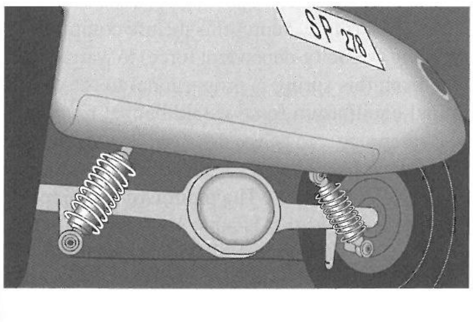
R0004E Modellering och reglering - Övningspass 1:

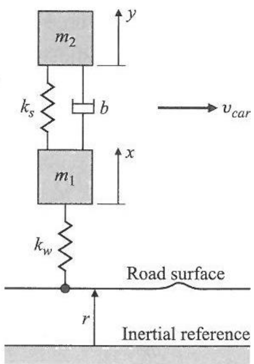
1. Write differential equations for the bridged tee circuit where v_i is the input and v_{C1} , v_{C2} are the states.



2. Write the equations of motion for the automobile and wheel motion assuming one-dimensional vertical motion of a mass above one wheel.



The quarter-car model



We use the following assumptions:

- force from the shock absorber is $\pm b(\dot{y} - \dot{x})$;
- gravitational forces are neglected;
- force in the wheel is $\pm k_w(x - r)$.

3. Determine the order of the following transfer functions:

$$\frac{X(s)}{F(s)} = \frac{7}{s^2 + 5s + 10}$$

$$\frac{X(s)}{F(s)} = \frac{15}{(s+10)(s+11)}$$

$$\frac{X(s)}{F(s)} = \frac{s+3}{s^3 + 11s^2 + 12s + 18}$$

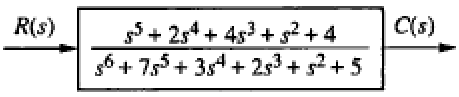


FIGURE P2.1

$$G(s) = \frac{5(s+15)(s+26)(s+72)}{s(s+55)(s^2+5s+30)(s+56)(s^2+27s+52)}$$

4. Determine the order of the following state space models:

a.

$$\mathbf{x} = \begin{bmatrix} -4 & -1.5 \\ 4 & 0 \end{bmatrix}\mathbf{x} + \begin{bmatrix} 2 \\ 0 \end{bmatrix}u(t)$$

$$y = [1.5 \quad 0.625]\mathbf{x}$$

b.

$$\begin{bmatrix} \dot{x}_1 \\ \dot{v}_1 \\ \dot{x}_2 \\ \dot{v}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ -K/M_1 & -D/M_1 & K/M_1 & 0 \\ 0 & 0 & 0 & 1 \\ K/M_2 & 0 & -K/M_2 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ v_1 \\ x_2 \\ v_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1/M_2 \end{bmatrix} f(t)$$

c.

$$\dot{\mathbf{z}} = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ -1 & -1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & -1 & -1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & -1 & -1 \end{bmatrix}\mathbf{z} + \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} f(t)$$

$$y = [\; 0 \quad 0 \quad 0 \quad 0 \quad 1 \quad 0]\mathbf{z}$$

d.

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 1.1 & 0.3 & -1.5 \\ 0.1 & 3.5 & 2.2 \\ 0.4 & 2.4 & -1.1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0.1 & 0 \\ 0 & 1.1 \\ 1.0 & 1.0 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$$

$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$$

5.

PROBLEM: Given the transfer function

$$G(s) = \frac{100}{s^2 + 15s + 100}$$

find T_p, $\%OS$, T_s, and T_r.